

How to get Datasets for Machine Learning

The key to success in the field of machine learning or to become a great data scientist is to practice with different types of datasets. But discovering a suitable dataset for each kind of machine learning project is a difficult task. So, in this topic, we will provide the detail of the sources from where you can easily get the dataset according to your project.

Before knowing the sources of the machine learning dataset, let's discuss datasets.

What is a Dataset?

A **dataset** is a collection of data in which data is arranged in some order. A dataset can contain any data from a series of an array to a database table. Below table shows an example of the dataset:

Country	Age	Salary	Purchased
India	38	48000	No
France	43	45000	Yes
Germany	30	54000	No
France	48	65000	No
Germany	40		Yes
India	35	58000	Yes

A tabular dataset can be understood as a database table or matrix, where each column corresponds to a **particular variable**, and each row corresponds to the **fields of the dataset**. The most supported file type for a tabular dataset is "**Comma Separated File**," or **CSV**. But to store a "tree-like data," we can use the JSON file more efficiently.

Types of data in datasets

- **Numerical data:** Such as house price, temperature, etc.
- **Categorical data:** Such as Yes/No, True/False, Blue/green, etc.
- **Ordinal data:** These data are similar to categorical data but can be measured on the basis of comparison.

Note: A real-world dataset is of huge size, which is difficult to manage and process at the initial level. Therefore, to practice machine learning algorithms, we can use any dummy dataset.

Need of Dataset

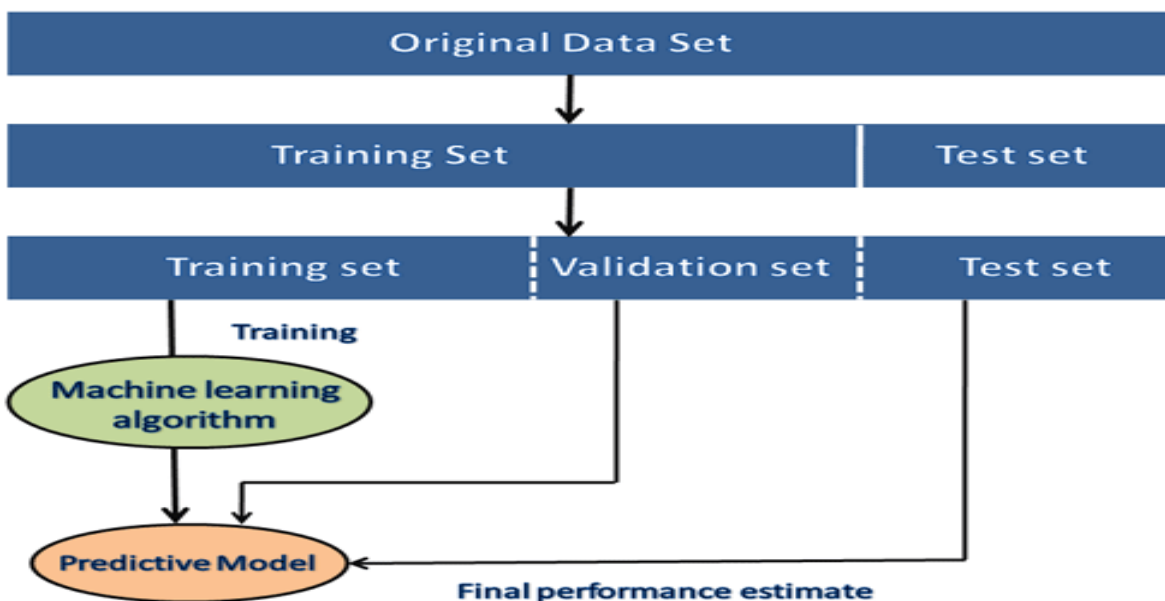
To work with machine learning projects, we need a huge amount of data, because, without the data, one cannot train ML/AI models. Collecting and preparing the dataset is one of the most crucial parts while creating an ML/AI project.

The technology applied behind any ML projects cannot work properly if the dataset is not well prepared and pre-processed.

During the development of the ML project, the developers completely rely on the datasets. In building ML applications, datasets are divided into two parts:

➤ **Training dataset:**

➤ **Test Dataset**

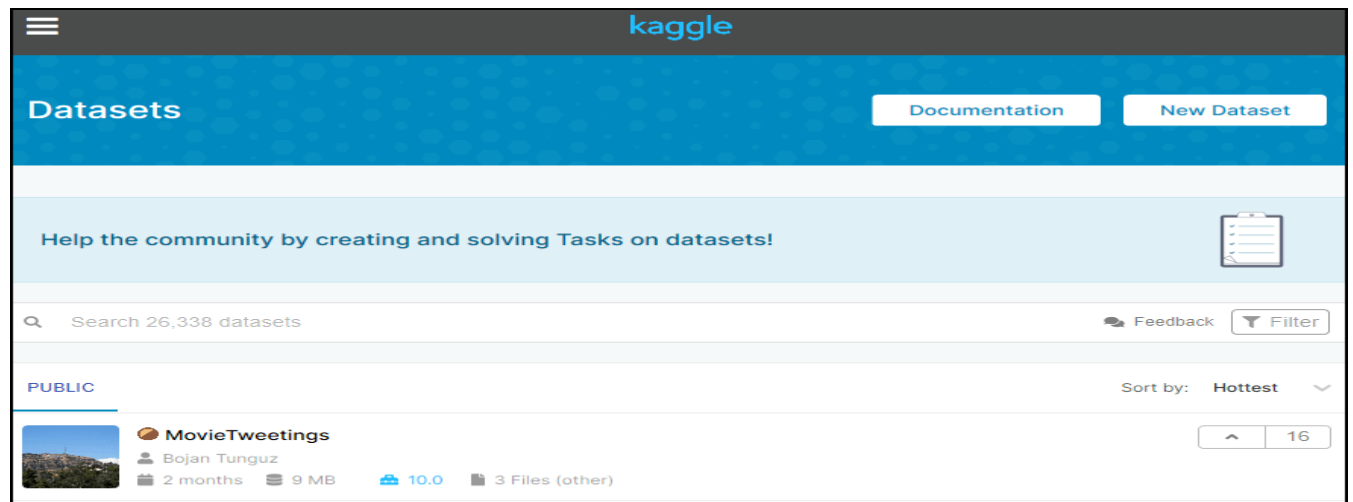


Note: The datasets are of large size, so to download these datasets, you must have fast internet on your computer.

Popular sources for Machine Learning datasets

The following is the list of datasets which are freely available for the public to work on it:

1. Kaggle Datasets



Kaggle is one of the best sources for providing datasets for Data Scientists and Machine Learners. It allows users to find, download, and publish datasets in an easy way. It also provides the opportunity to work with other machine learning engineers and solve difficult Data Science related tasks.

Kaggle provides a high-quality dataset in different formats that we can easily find and download.

The link for the Kaggle dataset is <https://www.kaggle.com/datasets>.

2. UCI Machine Learning Repository



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Default Task	Name	Data Types	Default Task	Attribute Types	# Instances	# Attributes	Year
Classification (360) Regression (107) Clustering (90) Other (55)	 Abalone	Multivariate	Classification	Categorical, Integer, Real	4177	8	1995
Attribute Type Categorical (38) Numerical (325) Mixed (55)	 Adult	Multivariate	Classification	Categorical, Integer	48842	14	1996
Data Type Multivariate (374) Univariate (24) Sequential (51) Time-Series (99) Text (55) Domain-Theory (23) Other (21)	 Annealing	Multivariate	Classification	Categorical, Integer, Real	798	38	
Area Life Sciences (110) Physical Sciences (52) CS / Engineering (479)	 Anonymous Microsoft Web Data		Recommender-Systems	Categorical	37711	294	1998
	 Arrhythmia	Multivariate	Classification	Categorical, Integer, Real	452	279	1998

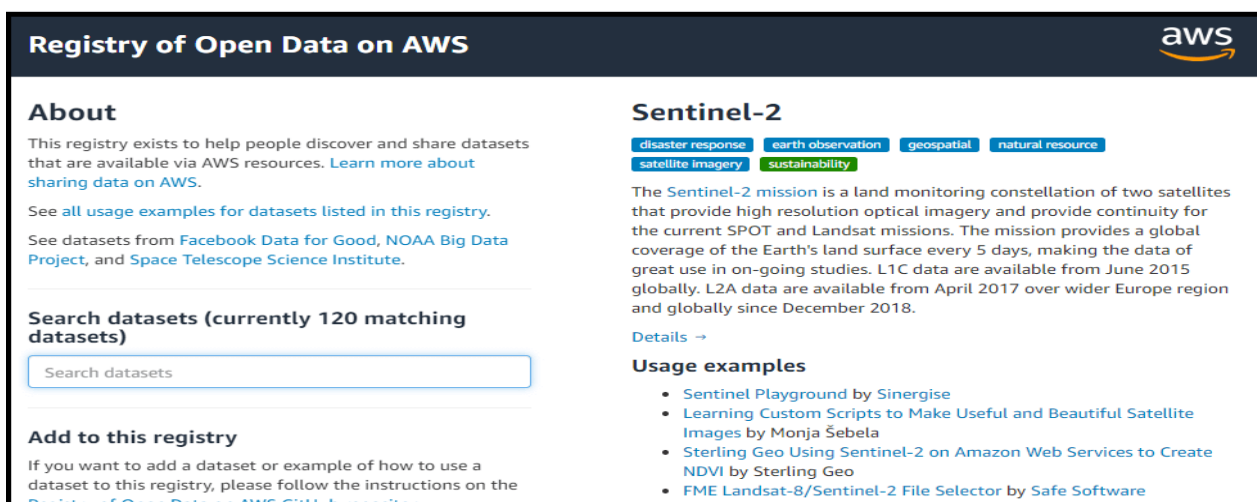
UCI Machine learning repository is one of the great sources of machine learning datasets. This repository contains databases, domain theories, and data generators that are widely used by the machine learning community for the analysis of ML algorithms.

Since the year 1987, it has been widely used by students, professors, researchers as a primary source of machine learning dataset.

It classifies the datasets as per the problems and tasks of machine learning such as **Regression, Classification, Clustering, etc.** It also contains some of the popular datasets such as the **Iris dataset, Car Evaluation dataset, Poker Hand dataset, etc.**

The link for the UCI machine learning repository is <https://archive.ics.uci.edu/ml/index.php>.

3. Datasets via AWS



The screenshot shows the 'Registry of Open Data on AWS' website. The header includes the AWS logo. The main content area is divided into two columns. The left column, titled 'About', contains text about the registry's purpose and links to learn more, usage examples, and datasets from various sources. It also features a search bar with the text 'Search datasets' and a note that there are currently 120 matching datasets. Below the search bar is a section titled 'Add to this registry' with instructions on how to contribute. The right column, titled 'Sentinel-2', features a list of tags (disaster response, earth observation, geospatial, natural resource, satellite imagery, sustainability) and a detailed description of the Sentinel-2 mission. It also includes a 'Details' link and a 'Usage examples' section with a list of links to various projects and scripts.

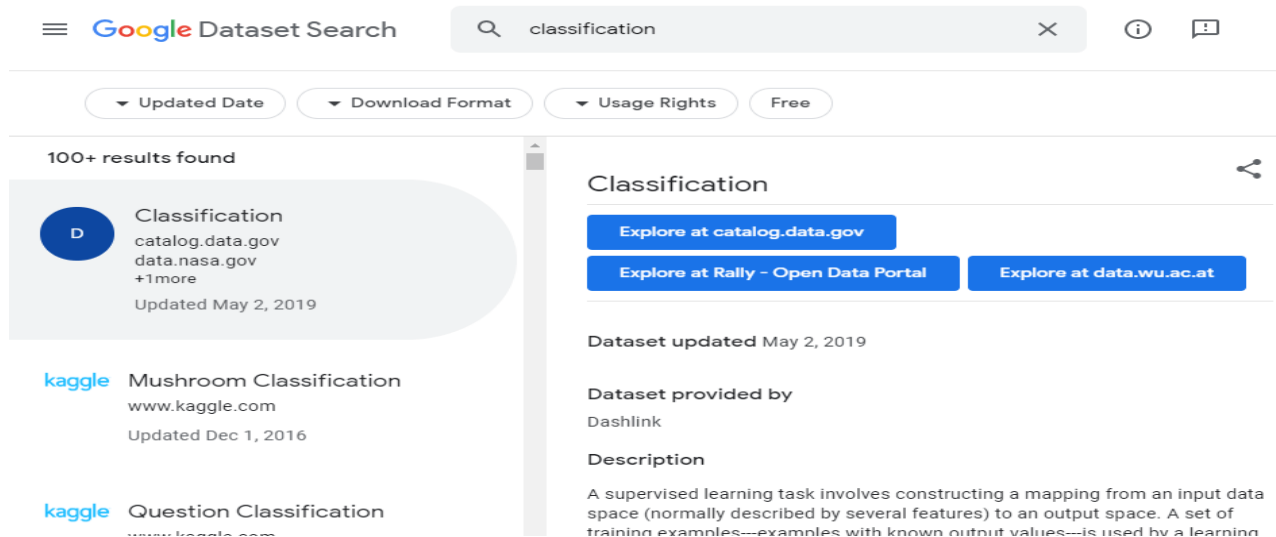
We can search, download, access, and share the datasets that are publicly available via AWS resources. These datasets can be accessed through AWS resources but provided and maintained by different government organizations, researches, businesses, or individuals.

Anyone can analyze and build various services using shared data via AWS resources. The shared dataset on cloud helps users to spend more time on data analysis rather than on acquisitions of data.

This source provides the various types of datasets with examples and ways to use the dataset. It also provides the search box using which we can search for the required dataset. Anyone can add any dataset or example to the **Registry of Open Data on AWS**.

The link for the resource is <https://registry.opendata.aws/>.

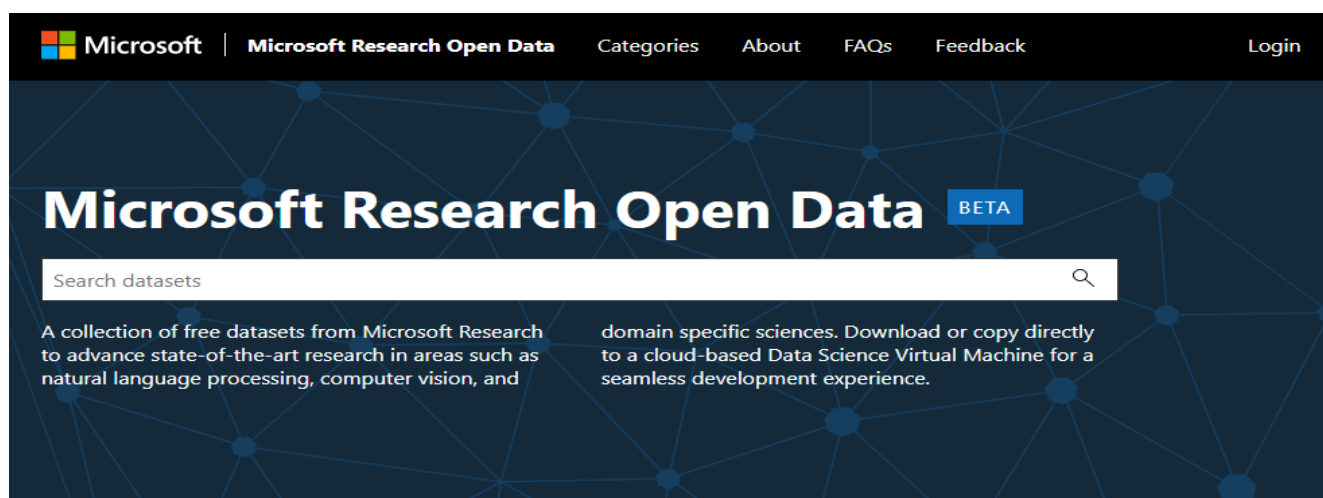
4. Google's Dataset Search Engine



Google dataset search engine is a search engine launched by **Google** on **September 5, 2018**. This source helps researchers to get online datasets that are freely available for use.

The link for the Google dataset search engine is <https://toolbox.google.com/datasetsearch>.

5. Microsoft Datasets



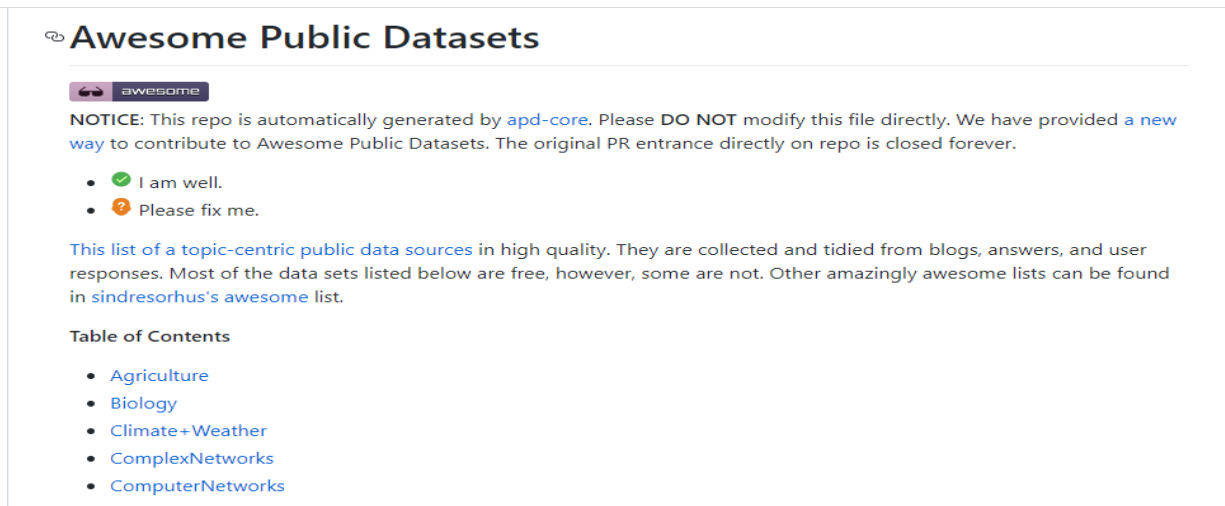
Dataset Categories

The Microsoft has launched the "**Microsoft Research Open data**" repository with the collection of free datasets in various areas such as **natural language processing, computer vision, and domain-specific sciences**.

Using this resource, we can download the datasets to use on the current device, or we can also directly use it on the cloud infrastructure.

The link to download or use the dataset from this resource is <https://msropendata.com/>.

6. Awesome Public Dataset Collection

The screenshot shows the top of a GitHub repository titled 'Awesome Public Datasets'. It includes a 'NOTICE' section stating the repository is auto-generated by 'apd-core' and should not be modified directly. Below this is a list of two items: a green checkmark icon with the text 'I am well.' and an orange question mark icon with the text 'Please fix me.'. A paragraph follows, explaining that the list contains high-quality, topic-centric public data sources collected from various sources. It mentions that most datasets are free but some are not, and refers to 'sindresorhus's awesome list' for more information. At the bottom, there is a 'Table of Contents' section with a bulleted list of topics: Agriculture, Biology, Climate+Weather, ComplexNetworks, and ComputerNetworks.

Awesome public dataset collection provides high-quality datasets that are arranged in a well-organized manner within a list according to topics such as Agriculture, Biology, Climate, Complex networks, etc. Most of the datasets are available free, but some may not, so it is better to check the license before downloading the dataset.

The link to download the dataset from Awesome public dataset collection is <https://github.com/awesomedata/awesome-public-datasets>.

7. Government Datasets

There are different sources to get government-related data. Various countries publish government data for public use collected by them from different departments.

The goal of providing these datasets is to increase transparency of government work among the people and to use the data in an innovative approach. Below are some links of government datasets:

- [Indian Government dataset](#)
- [US Government Dataset](#)
- [Northern Ireland Public Sector Datasets](#)
- [European Union Open Data Portal](#)

8. Computer Vision Datasets